

SUDIP KOIRALA

FULL STACK ENGINEER • AI DEVELOPER • SECURITY DEVELOPER
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PROFESSIONAL SUMMARY

Cybersecurity-trained full-stack software engineer with hands-on experience building secure, scalable systems end-to-end from architecture through deployment. Proven ability to work independently: implementing AES/RSA encryption systems, building file encryption tools, developing autonomous AI agents with OpenClaw, and shipping full-stack applications with Node.js, Next.js, SQL Server, MySQL, and Redis. Runs a personal security homelab covering SIEM/SOC telemetry (Wazuh, ELK, Splunk), Active Directory attack-and-defend scenarios, and malware analysis with YARA detection engineering. Strong foundation in networking, cryptography, and secure system design, currently applied to AI infrastructure, training data quality, and agentic systems. Seeking to bring this combination of security depth and full-stack engineering to a high-scale product team.

SKILLS

Languages & Development

Python • Node.js • SQL • REST APIs • Full-Stack Development • Git

Security

Cybersecurity • Cryptography • AES • Penetration Testing • Active Directory Security

Security Operations & Detection

SIEM (Wazuh, Splunk, ELK Stack) • SOC Monitoring & Alert Triage • Log Analysis • Detection Engineering • MITRE ATT&CK • Malware Analysis • YARA

AI & Automation

Agentic AI Development • Prompt Engineering • Large Language Models (LLM) • OpenClaw

Infrastructure & Tools

Linux • Red Hat Linux • Redis • Infrastructure as Code (IaC) • Virtualization • Remote Collaboration

WORK HISTORY

Freelance Software Engineer & Security Developer

2018 – Present

Self-Employed - Independent Projects, Remote

- Independently designed, built, and shipped software projects end-to-end - from architecture through deployment spanning full-stack web applications, security tooling, and AI systems.
- Built file encryption tools using AES-256 and RSA public-key cryptography in Python, ensuring secure data storage and transmission.
- Designed and deployed full-stack web applications using Next.js and Node.js with SQL Server and MySQL backends, including authentication and session management.
- Implemented Redis caching layers to optimize application performance and reduce database load.
- Developed and deployed AI agents using OpenClaw, automating complex workflows and multi-step reasoning tasks.
- Applied networking fundamentals (TCP/IP, DNS, firewalls, VPNs) to design secure application architectures.
- Built Python-based automation scripts and AI pipelines integrating large language models for data processing tasks.
- Conducted security assessments and implemented encryption best practices across web and file-based systems.
- Collaborated directly with clients to design, develop, and publish a mobile app named "Clef" on the Google Play Store and a business website for a beauty salon.

Back-End Web Developer

Jan 2017 – Jun 2017

Broadway Infosys Nepal - Kathmandu, Central Development Region, Nepal

- Worked as a junior web developer using PHP, Java, MySQL, HTML, and JavaScript.
- Provided ongoing maintenance, bug fixes, and patch sets for existing web and desktop applications.
- Gained hands-on experience across all phases of the software development lifecycle, applying strong knowledge of algorithms and data structures.

SECURITY HOMELAB EXPERIENCE

SIEM & SOC Telemetry Lab (Wazuh / ELK / Splunk)

- Built a virtualized SOC environment that centralizes endpoint and network telemetry from Windows and Linux hosts into Wazuh, the Elastic Stack (ELK), and Splunk, allowing side-by-side comparison of SIEM platforms.
- Deployed Wazuh agents and Sysmon for endpoint visibility and onboarded Windows Event Logs, Linux syslog/auth logs, and network logs through Logstash and Splunk forwarders.
- Wrote custom detection rules, correlation searches, and dashboards; validated alerting by simulating adversary activity and mapping detections to MITRE ATT&CK techniques.
- Practiced SOC analyst workflows: alert triage, log investigation, false-positive tuning, and documenting findings.

Tech: Wazuh, Elasticsearch, Logstash, Kibana, Splunk, Sysmon, MITRE ATT&CK

Active Directory Attack-and-Defend Lab (GOAD)

- Deployed GOAD (Game of Active Directory), a multi-domain, multi-forest vulnerable Windows environment, to practice both offensive and defensive Active Directory security.
- Executed common AD attack paths including enumeration, Kerberoasting, AS-REP roasting, password spraying, NTLM relay, ACL abuse, and lateral movement using BloodHound and Impacket.
- Forwarded domain controller and host logs into the SIEM lab to build and test detections for each technique, then applied hardening measures to close the identified attack paths.

Tech: GOAD, Active Directory, Windows Server, Kerberos, BloodHound, Impacket

Malware Analysis & YARA Rule Generation

- Built an isolated, snapshot-based analysis environment for safely handling malware samples with no access to production networks.
- Performed static analysis (file hashing, PE header and import inspection, string extraction) and dynamic analysis (process, file system, registry, and network behavior) to extract indicators of compromise.
- Generated and authored YARA rules from extracted strings, byte patterns, and file characteristics, then tested and tuned them against sample sets to improve detection and reduce false positives.

Tech: YARA, Python, PE Analysis, FLARE-VM, REMnux

PROJECTS

FIFA Tournament Manager - Full Stack Web Application

GitHub: github.com/siuee/Tournament | Live: tournament-orcin.vercel.app

Built a full-featured tournament management platform for FIFA players. Users create player cards with stats, organize 1v1 and 2v2 tournaments with auto-generated fixtures, and view broadcast-style match previews. Includes a live betting system with form-adjusted odds, automatic bet settlement, Ballon d'Or rankings, and per-player stat tracking.

Tech: Node.js, Firebase

File Encryption Tool

A Python-based file encryption application using AES-256 and RSA key exchange. Supports secure key generation, file encryption/decryption, and integrity verification via SHA-256 hashing.

Tech: Python, PyCryptodome, RSA, AES

Full Stack Web App with Secure Auth

Built a full stack application with a Next.js frontend and Node.js backend, featuring JWT authentication, Redis session management, and MySQL data persistence, plus rate limiting and input sanitization.

Tech: Next.js, Node.js, MySQL, Redis, JWT

Nepal Import Analytics - Data Visualization Platform

GitHub: github.com/siuee/nepalImports | Live: nepal-imports.vercel.app

Built a data-driven platform analyzing import expenditure across product categories in Nepal, with actionable agricultural insights (cultivation guidance, recommended growing locations) to reduce import dependency - combining economic data visualization with policy-relevant recommendations.

Tech: Next.js, Vercel

OpenClaw AI Agent - Workflow Automation

Designed and deployed an AI agent using OpenClaw capable of multi-step task execution, tool use, and autonomous decision-making for workflow automation use cases.

Tech: OpenClaw, Python, LLMs

EDUCATION

B.S. in Cybersecurity

Texas A&M University-Commerce - Commerce, TX

Graduated 2026